

IKHLAQ ZAHIR

COMPUTER SCIENCE GRADUATE,
GAME DEVELOPMENT, HCI & AI-DRIVEN
INTERACTIVE SYSTEMS



CONTACT

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Google Play (Developer):
play.google.com/store/apps/developer?id=IsolatedVertex

EDUCATION

**Bachelors of Science in
Computer Science**
University of Management
and Technology, Lahore
2024

**Intermediate (pre-
engineering)**
Aga Khan Higher Secondary
School, Gilgit
2019

Matriculation
Aga Khan Higher Secondary
School, Gahkuch
2017

SUMMARY

Computer Science graduate with experience in independently designing, developing, testing, and deploying games and Android applications. Development background spans Unity/C#, Kotlin/Android, algorithm design, procedural content generation. Seeking graduate research opportunities combining a software and game-development background with formal research in game technology, HCI, AI, interactive systems, or educational games.

CORE COMPETENCIES

Game Development:

- Unity Engine, C#, 2D/3D Development
- Game Mechanics & Systems Design
- Game Architecture, Procedural Content Generation
- Constraint-Based Puzzle Design

Android Development:

- Kotlin, Android SDK, Jetpack Compose
- Android UI/UX, SQLite, JSON Processing
- In-App Billing, Google Play Deployment
- Mobile Application Architecture

AI & Machine Learning:

- Python, PyTorch, Machine Learning Fundamentals
- Generative AI, Large Language Models (LLMs)
- Transformer-Based Models, AI-Assisted Development
- On-Device/Lightweight AI

HCI & Interactive Systems:

- User Interface Design, UX-Oriented Development
- Interactive System Design, Player Experience
- Human-Computer Interaction, Interactive Learning Systems
- Game-Based Learning, User-Centered Design

TOOLS AND
PLATFORMS

- Visual Studio
- VS Code
- Android Studio
- Unity
- Gradle
- SQLite

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ADDITIONAL INFO

Programming Languages:

C#, Kotlin, Python,
JavaScript/TypeScript

Game Engines:

Unity, Android Studio

Mobile Platform:

Android

AI/ML Tools:

PyTorch, Hugging Face, Google
Colab

Languages Spoken:

English, Urdu

PROFESSIONAL EXPERIENCE

Independent Game & Application Developer (2024 – current)

- Independently designed and developed multiple games and Android applications from concept through implementation, testing, iteration, and Google Play deployment.
- Game and application architecture; gameplay and interaction design.
- Programming, debugging, and algorithm development.
- UI/UX implementation and procedural generation.
- AI and gameplay systems; Android performance optimization.
- Database and data management.
- Testing, iteration, and Google Play release management.

CERTIFICATIONS

Android Game Development
in Unity 3D – PNY Trainings

May 2022

Unity 3D Game Design and
Development – Punjab Board
of Technical Education Lahore

October 2023

PROFESSIONAL EXPERIENCE

- Game Development, Game Architecture, and Game Technology
- Game AI and Intelligent Interactive Systems
- Human-Computer Interaction (HCI) and User Experience
- Procedural Content Generation Level Design and environment Design
- Algorithmic Game Systems
- Educational and Serious Games

- Human-AI Interaction
- Interactive Learning Systems
- XR, Immersive Interaction, and Interactive Experiences
- Computational Game Design and Game Architecture

RESEARCH DIRECTIONS

Game Based Learning:

How interactive game mechanics can communicate complex scientific and technical concepts more effectively than conventional information presentation

Intelligent Game Systems:

How AI can adapt game experiences according to player behavior, knowledge, engagement, and ability

Procedural Content Generation:

How algorithms can generate meaningful game content while maintaining constraints, coherence, difficulty, and player experience

Human – AI Interaction in Games:

How AI agents can become interactive teachers, companions, NPCs, or collaborators within game environments

Interactive Simulations:

HCI and Immersive Experiences:

How novel interaction technologies, XR, and interactive environments can create more engaging and effective learning and gaming experiences

RESEARCH SKILLS

- Independent problem solving and algorithmic thinking.
- Technical prototyping and system design.
- Experimental thinking and iterative development.
- Translating concepts into working prototypes.
- Cross-disciplinary thinking spanning computing, games, AI, and HCI.